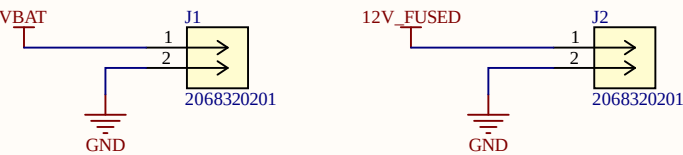
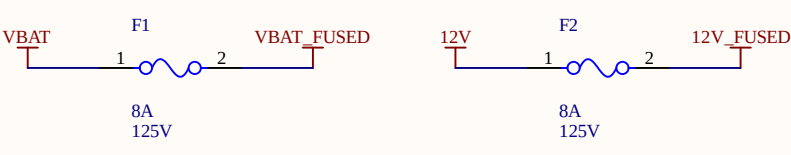


CONNECTORS

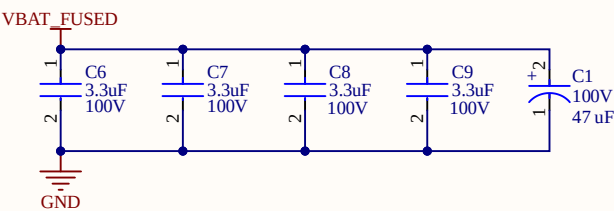


FUSE HOLDERS

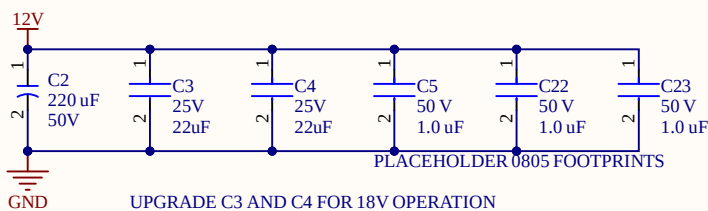


FUSES MUST BE RATED FOR 12A ON OUTPUT, 5A ON INPUT
INPUT CURRENT IS ~2.5A AT MAXIMUM LOAD CURRENT OF 9A, ASSUMING 90% EFFICIENCY

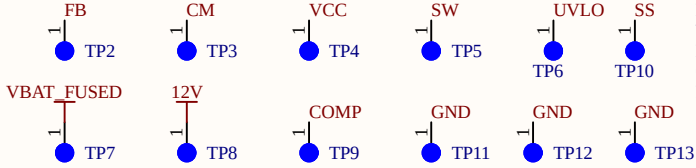
INPUT FILTERING



OUTPUT FILTERING

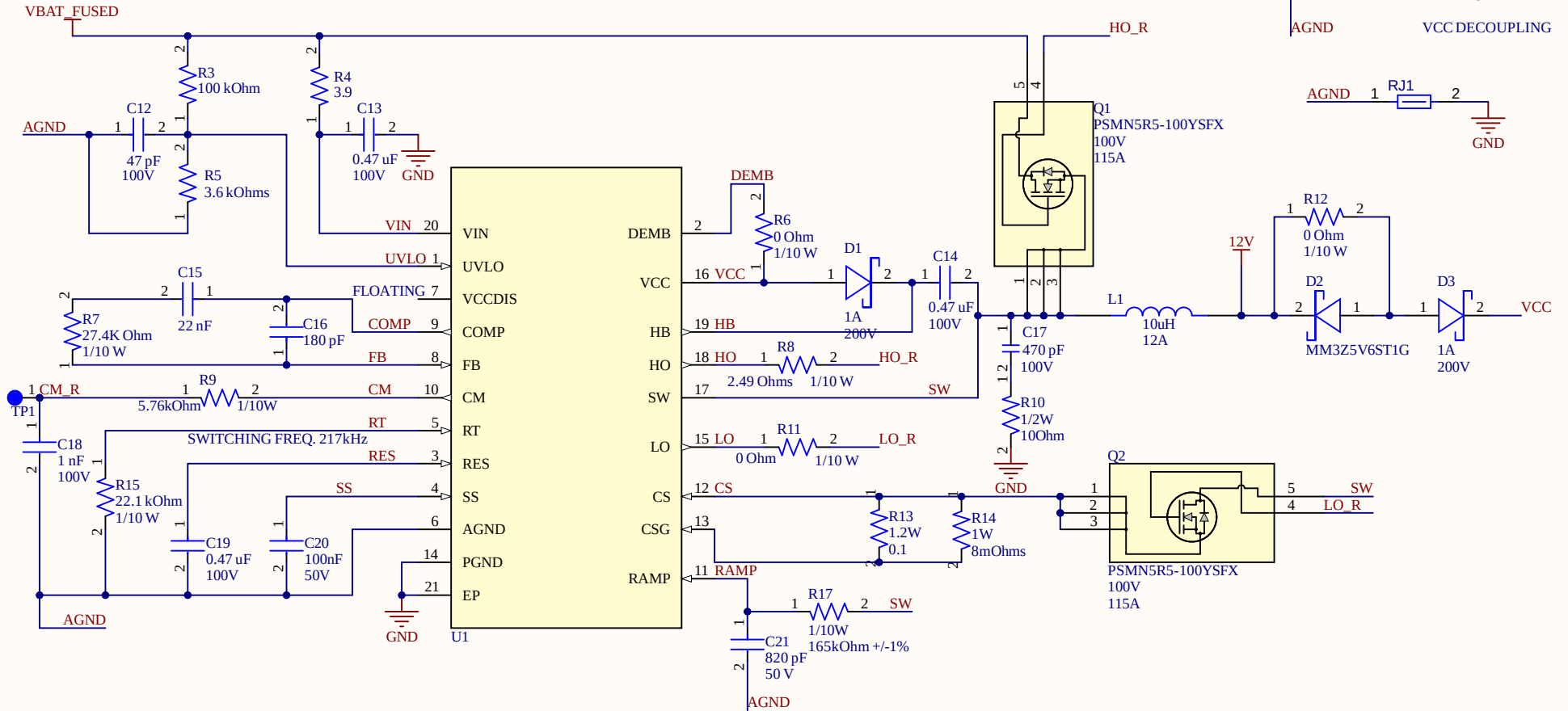


UPGRADE C3 AND C4 FOR 18V OPERATION



UVLO SIZED FOR 36V TO ACCOMMODATE BATTERY VOLTAGE CHANGES, 2V HYSTERESIS ON BOOT
 VIN HAS LPF WITH R4 AND C13, CUTOFF ~87kHz TO ELIMINATE SWITCHING FREQ
 VCCDIS HAS INTERNAL PULLDOWN TO ENABLE INTERNAL LDO/EXTERNAL 12V ONCE STABLE
 COMP-FB CIRCUIT IS USED TO ENSURE BUCK RESPONDS TO DC CHANGES IN OUTPUT, DOESN'T OVERCORRECT TO TRANSIENTS
 CM (CURRENT MONITOR) HAS LPF TO CUTOFF SWITCHING NOISE
 RES & SS DETERMINE HICCUP PERIOD & RISE TIME

DEMB- WHEN LOW, EFFICIENCY BETTER, BUT WORSE TRANSIENT RESPONSE (SWITCHING FREQ. VARIES)
 VCC DRIVES FET GATES, DIODE SO NO CURRENT FLOWS INTO VOUT
 HO/LO HAS DAMPING RESISTORS TO CREATE RLC DUE TO HIGH SWITCHING
 CS (CURRENT SENSE) MEASURES WHEN LO SIDE IS ON, TRIPS AT 16.4A (7.3mOhms Req)
 RAMP PIN USED TO EMULATE L CURRENT USING VOLTAGE ON EITHER SIDE OF L



FB = 0.8V @12V

VCC DECOUPLING